Blackfin Families:

BF54x: the most appropriate family for our system, the table below shows bf54x peripherals, I think we can use BF548 or BF549

Processor Features		ADSP-BF549	ADSP-BF548	ADSP-BF547	ADSP-BF544	ADSP-BF542
Lockbox® [1] code security		1	1	1	1	1
128-bit AES/ ARC4 data encryption		1	1	1	1	1
SD/SDIO controller		1	1	1	–	1
Pixel compositor		1	1	1	1	1
18- or 24-bit EPPI0 with LCD		1	1	1	1	–
16-bit EPPI1, 8-bit EPPI2		1	1	1	1	1
Host DMA port		1	1	1	1	–
NAND flash controller		1	1	1	1	1
ATAPI		1	1	1	–	1
High speed USB OTG		1	1	1	–	1
Keypad interface		1	1	1	–	1
MXVR		1	–	–	–	–
CAN ports		2	2	–	2	1
TWI ports		2	2	2	2	1
SPI ports		3	3	3	2	2
UART ports		4	4	4	3	3
SPORTs		4	4	4	3	3
Up/down counter		1	1	1	1	1
Timers		11	11	11	11	8
General-purpose I/O pins		152	152	152	152	152
Memory Configura-tions (K Bytes)	L1 Instruction SRAM/cache	16	16	16	16	16
	L1 Instruction SRAM	48	48	48	48	48
	L1 Data SRAM/cache	32	32	32	32	32
	L1 Data SRAM	32	32	32	32	32
	L1 Scratchpad SRAM	4	4	4	4	4
	L1 ROM[2]	64	64	64	64	64
	L2	128	128	128	64	–
	L3 Boot ROM[2]	4	4	4	4	4
Maximum core instruction rate (MHz)		533	533	600	533	600

[1] Lockbox is a registered trademark of Analog Devices, Inc.
[2] This ROM is not customer-configurable.

BF52x:

Feature	ADSP-BF522	ADSP-BF524	ADSP-BF526	ADSP-BF523	ADSP-BF525	ADSP-BF527
Host DMA	1	1	1	1	1	1
USB	–	1	1	–	1	1
Ethernet MAC	–	–	1	–	–	1
Internal Voltage Regulator	–	–	–	1	1	1
TWI	1	1	1	1	1	1
SPORTs	2	2	2	2	2	2
UARTs	2	2	2	2	2	2
SPI	1	1	1	1	1	1
GP Timers	8	8	8	8	8	8
GP Counter	1	1	1	1	1	1
Watchdog Timers	1	1	1	1	1	1
RTC	1	1	1	1	1	1
Parallel Peripheral Interface	1	1	1	1	1	1
GPIOs	48	48	48	48	48	48
Memory (bytes) L1 Instruction SRAM	48K	48K	48K	48K	48K	48K
Memory (bytes) L1 Instruction SRAM/Cache	16K	16K	16K	16K	16K	16K
Memory (bytes) L1 Data SRAM	32K	32K	32K	32K	32K	32K
Memory (bytes) L1 Data SRAM/Cache	32K	32K	32K	32K	32K	32K
Memory (bytes) L1 Scratchpad	4K	4K	4K	4K	4K	4K
Memory (bytes) L3 Boot ROM	32K	32K	32K	32K	32K	32K
Maximum Instruction Rate[1]	400 MHz			600 MHz		
Maximum System Clock Speed	100 MHz			133 MHz		
Package Options	289-Ball CSP_BGA					
	208-Ball CSP_BGA					

[1] Maximum instruction rate is not available with every possible SCLK selection.

The ADSP-BF60x processors are the Dual-Core processor up to 1GHz with Hardware Support for HD Video Analytic's

Table 1. Processor Comparison

Processor Feature	ADSP-BF606	ADSP-BF607	ADSP-BF608	ADSP-BF609
Up/Down/Rotary Counters	1			
Timer/Counters with PWM	8			
3-Phase PWM Units (4-pair)	2			
SPORTs	3			
SPIs	2			
USB OTG	1			
Parallel Peripheral Interface	3			
Removable Storage Interface	1			
CAN	1			
TWI	2			
UART	2			
ADC Control Module (ACM)	1			
Link Ports	4			
Ethernet MAC (IEEE 1588)	2			
Pixel Compositor (PIXC)	No		1	1
Pipelined Vision Processor (PVP) Video Resolution[1]	No		VGA	HD
Maximum PVP Line Buffer Size	N/A		640	1280
GPIOs	112			

The ADSP-BF70x processors are the latest Low Power 400MHz Blackfin+ embedded processor with 1MByte L2 SRAM & DDR2/LPDDR Interface

Processor Feature	ADSP-BF700	ADSP-BF701	ADSP-BF702	ADSP-BF703	ADSP-BF704	ADSP-BF705	ADSP-BF706	ADSP-BF707
Maximum Speed Grade (MHz)[1]	200		400					
Maximum SYSCLK (MHz)	100		200					
Package Options	88-Lead LFCSP	184-Ball CSP_BGA	88-Lead LFCSP	184-Ball CSP_BGA	88-Lead LFCSP	184-Ball CSP_BGA	88-Lead LFCSP	184-Ball CSP_BGA
GPIOs	43	47	43	47	43	47	43	47
Memory (bytes) L1 Instruction SRAM	48K							
Memory (bytes) L1 Instruction SRAM/Cache	16K							
Memory (bytes) L1 Data SRAM	32K							
Memory (bytes) L1 Data SRAM/Cache	32K							
Memory (bytes) L1 Scratchpad (L1 Data C)	8K							
Memory (bytes) L2 SRAM	128K		256K		512K		1024K	
Memory (bytes) L2 ROM	512K							
Memory (bytes) DDR2/LPDDR (16-bit)	No	Yes	No	Yes	No	Yes	No	Yes
I^2C	1							
Up/Down/Rotary Counter	1							
GP Timer	8							
Watchdog Timer	1							
GP Counter	1							
SPORTs	2							
Quad SPI	2							
Dual SPI	1							
SPI Host Port	1							
USB 2.0 HS OTG	1							
Parallel Peripheral Interface	1							
CAN	2							
UART	2							
Real-Time Clock	1							
Static Memory Controller (SMC)	Yes							
Security Crypto Engine	Yes							
SD/SDIO (MSI)	4-bit	8-bit	4-bit	8-bit	4-bit	8-bit	4-bit	8-bit
4-Channel 12-Bit ADC	No	Yes	No	Yes	No	Yes	No	Yes

[1] Other speed grades available.